

Bone Fracture Classification Using EfficientNetB4

A Deep Learning Approach to Automated X-ray Image Classification

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Live Application: <https://bonescanai.streamlit.app>

Repository: <https://github.com/Fiazbkh/Bone-Fracture-Classification-Using-Deep-Learning>

Abstract

This project presents a deep learning pipeline for the automated detection of bone fractures from multi-region X-ray images. The system employs EfficientNetB4, a convolutional neural network pretrained on the ImageNet dataset, through transfer learning. The model was trained on a binary classification dataset containing 9,246 X-ray images across multiple anatomical regions, with validation and test sets of 829 and 506 images respectively. By applying EfficientNetB4-specific preprocessing and a structured training strategy, the model achieved a test accuracy of 98.02 percent, an F1 score of 0.98, and an AUC score of 0.9985. The trained model is deployed as an interactive web application using Streamlit, enabling real-time fracture detection from uploaded X-ray images.

1. Introduction

Bone fractures are among the most common injuries encountered in clinical practice. Timely and accurate diagnosis through radiographic examination is essential for effective patient treatment. Manual interpretation of X-ray images is time-consuming, subject to observer variability, and requires significant radiological expertise. In resource-limited environments and high-volume clinical settings, these constraints can lead to diagnostic delays and errors.

Advances in deep learning, particularly in convolutional neural networks, have demonstrated strong potential for automated medical image analysis. Models such as EfficientNetB4, which balance accuracy and computational efficiency through compound scaling, have achieved state-of-the-art results across image classification tasks. When combined with transfer learning, these models can be adapted to medical imaging domains with relatively modest amounts of labeled data.

This project applies EfficientNetB4 to bone fracture detection using a publicly available multi-region X-ray dataset. The objective is to develop a model capable of accurately distinguishing fractured from non-fractured bone X-rays across multiple anatomical regions and to deploy that model as a practical web-based tool.

1.1 Objectives

- To build a binary classification model for bone fracture detection from X-ray images
- To apply transfer learning using EfficientNetB4 pretrained on ImageNet
- To evaluate model performance using accuracy, precision, recall, F1 score, and AUC
- To deploy the trained model as a Streamlit web application for real-time inference
- To document the complete pipeline for reproducibility and academic review

2. Related Work

The application of convolutional neural networks to medical image classification has been an active area of research. Studies have demonstrated that deep learning models can match or exceed radiologist-level performance on specific diagnostic tasks. Early work by Rajpurkar et al. introduced CheXNet, a model achieving radiologist-level pneumonia detection from chest X-rays. Subsequent work extended these methods to musculoskeletal imaging and fracture detection.

Transfer learning has emerged as a particularly effective strategy for medical imaging tasks where labeled data is limited. Models initialized with ImageNet weights consistently outperform those trained from scratch on medical imaging benchmarks. EfficientNet, introduced by Tan and Le in 2019, uses principled compound scaling to uniformly scale model depth, width, and resolution. EfficientNetB4 offers a strong balance between parameter efficiency and representational capacity, and has been applied successfully in skin lesion classification and retinal disease detection.

3. Dataset

3.1 Dataset Description

The dataset used is the Bone Fracture Multi-Region X-ray Dataset, publicly available through Kaggle. It contains X-ray images from multiple anatomical regions including the wrist, elbow, hand, shoulder, finger, and forearm. Each image is labeled as either fractured or not fractured.

3.2 Dataset Statistics

Split	Fractured	Not Fractured	Total
Train	4,606	4,640	9,246
Validation	337	492	829
Test	238	268	506
Total	5,181	5,400	10,581

The dataset is well balanced across both classes, which reduces the risk of class bias during training.

3.3 Data Preprocessing

All images were resized to 380 by 380 pixels, the default input resolution for EfficientNetB4. The EfficientNetB4-specific preprocessing function was applied to all images rather than simple pixel rescaling. This function normalizes pixel values to the range expected by the pretrained model.

Using standard 0-to-1 rescaling was found to reduce validation accuracy to approximately 65 percent, confirming the critical importance of matching preprocessing to the specific architecture.

3.4 Data Augmentation

Data augmentation was applied exclusively to the training set to improve model generalization:

- Horizontal flipping to account for bilateral symmetry in anatomical structures
- Random rotation within a 10-degree range to simulate variation in patient positioning
- Zoom variation of up to 10 percent to handle differences in image scale
- Width and height shifts of up to 10 percent to account for variable image framing

4. Methodology

4.1 Model Architecture

The model uses EfficientNetB4 with a custom classification head. The base model was loaded with pretrained ImageNet weights with the original top classification layer excluded. The following layers were added:

- Global Average Pooling to reduce spatial feature maps to a fixed-size vector
- Batch Normalization to stabilize and accelerate training
- Dropout with a rate of 0.4 to reduce overfitting
- A fully connected Dense layer with 256 units and ReLU activation
- A second Dropout layer with a rate of 0.3
- A final Dense layer with a single unit and sigmoid activation for binary output

The total model contains approximately 18.14 million parameters. During Phase 1, only 462,849 parameters in the classification head are trainable, while the 17.67 million base model parameters remain frozen.

4.2 Training Strategy

Phase 1: Top Layer Training

In Phase 1, the EfficientNetB4 base was frozen and only the custom layers were trained using the Adam optimizer at a learning rate of 0.001 with binary cross-entropy loss. This allows the classification head to adapt to the fracture detection task while preserving pretrained representations.

Training Callbacks

- EarlyStopping monitored validation accuracy with a patience of 4 epochs and restored best weights

- ReduceLROnPlateau reduced the learning rate by a factor of 0.3 when validation loss plateaued
- ModelCheckpoint saved the model achieving the best validation accuracy

4.3 Implementation Environment

The pipeline was implemented in Python using TensorFlow 2.20.0 and Keras, trained on Google Colab with an NVIDIA T4 GPU. The Kaggle API was used to download the dataset. The model was saved in Keras native format and deployed using Streamlit.

5. Results

5.1 Training Performance

Training ran for 10 epochs. The model showed strong improvement from the first epoch, achieving 93.49 percent validation accuracy in epoch 1 and reaching its best validation accuracy of 98.67 percent in epoch 9.

Epoch	Train Accuracy	Val Accuracy	Train Loss	Val Loss
1	88.72%	93.49%	0.3064	0.1826
2	94.40%	96.62%	0.1528	0.0938
5	97.27%	98.07%	0.0737	0.0514
9 (Best)	98.52%	98.67%	0.0438	0.0360

5.2 Test Set Evaluation

Metric	Fractured	Not Fractured	Overall
Precision	0.97	0.98	0.98
Recall	0.98	0.98	0.98
F1 Score	0.98	0.98	0.98
Accuracy	-	-	98.02%
AUC Score	-	-	0.9985
Correct / Total	-	-	496 / 506

5.3 Confusion Matrix Analysis

Out of 506 test images, 496 were correctly classified and only 10 were misclassified. Of 238 fractured images, 234 were correctly identified and 4 were missed. Of 268 non-fractured images, 262 were correct and 6 were false positives. The low false negative rate is clinically important, as missing a fracture carries greater risk than a false positive.

5.4 ROC Curve Analysis

The AUC score of 0.9985 indicates near-perfect discriminatory ability. The ROC curve rises sharply toward the top-left corner, confirming very high true positive rates across a wide range of classification thresholds with minimal false positives.

6. Web Application Deployment

The trained model is deployed as a web application at <https://bonescanai.streamlit.app> using the Streamlit framework. Users can upload X-ray images and receive instant fracture detection predictions without any technical knowledge or local installation. The application applies the same EfficientNetB4 preprocessing pipeline used during training before generating predictions and displays the result with a confidence score.

7. Discussion

The results demonstrate that transfer learning with EfficientNetB4 is highly effective for multi-region bone fracture detection. The model achieved near-perfect performance across all evaluation metrics. A critical finding was the sensitivity of EfficientNetB4 to preprocessing. Using standard pixel rescaling degraded validation accuracy to 65 to 70 percent, while the architecture-specific preprocessing immediately resolved this and enabled convergence to high accuracy within the first epoch. This emphasizes the importance of matching preprocessing to the specific architecture and pretraining configuration.

7.1 Limitations

- The dataset lacks metadata such as patient demographics or specific anatomical region labels
- The model was not evaluated against radiologist assessments on the same images
- The web application does not include uncertainty quantification for clinical use

8. Conclusion

This project demonstrates the development and deployment of a deep learning system for automated bone fracture detection. Using EfficientNetB4 with transfer learning and correct preprocessing, the model achieved 98.02 percent test accuracy, an F1 score of 0.98, and an AUC score of 0.9985. The model is deployed as a publicly accessible web application. Future work could extend this system to multi-class fracture severity classification and incorporate explainability methods such as Grad-CAM to highlight image regions contributing to each prediction.